Chapter 8

MALMQUIST PRODUCTIVITY INDEX

Efficiency Change Over Time

Kaoru Tone

National Graduate Institute for Policy Studies, 2-2 Wakamatsu, Shinjuku, Tokyo 162-8677, Japan email: tone@grips.ac.jp

Abstract: The Malmquist index (MI) evaluates the efficiency change over time. In the non-parametric framework, it is measured as the product of catch-up (or recovery) and frontier-shift (or innovation) terms, both coming from the DEA technologies. We introduce three different approaches for measuring the Malmquist index along with scale efficiency related subjects.

Key words: Malmquist index, catch-up, recovery, frontier-shift, innovation, radial, non-radial, slacks-based measure, oriented, non-oriented

1. INTRODUCTION

The concept of Malmquist productivity index was first introduced by Malmquist (1953), and has further been studied and developed in the non-parametric framework by several authors. See for example, among others, Caves, Christensen and Diewert (1982), Färe and Grosskopf (1992), Färe, Grosskopf, Lindgren and Roos (1989, 1994), Färe, Grosskopf and Russell (1998b) and Thrall (2000). It is an index representing Total Factor Productivity (TFP) growth of a Decision Making Unit (DMU), in that it reflects progress or regress in efficiency along with progress or regress of the frontier technology over time under the multiple inputs and multiple outputs framework. In this chapter we first introduce the concept of Malmquist productivity index, and then present three different approaches for its measurement: radial, non-radial, and non-radial and non-oriented approaches. Scale efficiency change is discussed in Section 4. We demonstrate with numerical examples for comparing different approaches in Section 5. Concluding remarks follow in Section 6.

2. DEALING WITH PANEL DATA

The Malmquist index evaluates the productivity change of a DMU between two time periods. It is defined as the product of "Catch-up" and "Frontier-shift" terms. The catch-up (or recovery) term relates to the degree that a DMU attains for improving its efficiency, while the frontier-shift (or innovation) term reflects the change in the efficient frontiers surrounding the DMU between the two time periods. We deal with a set of n DMUs (x_j, y_j) (j = 1,...n) each having m inputs denoted by a vector $x_j \in R^m$ and q outputs denoted by a vector $y_j \in R^q$ over the periods 1 and 2. We assume $x_j > 0$ and $y_j > 0\ (\forall j)$. The notations $(x_o, y_o)^1$ and $(x_o, y_o)^2$ are employed for designating DMU_o (o = 1, ..., n) in the periods 1 and 2 respectively.

The production possibility set $(X,Y)^t$ (t = 1 and 2) spanned by $\{(x_j, y_j)^t\}$ (j = 1, ..., n) is defined by:

$$(X,Y)^t = \left\{(x,y) \,\middle|\, x \ge \sum_{j=1}^{n} \lambda_j x_j^t,\ 0 \le y \le \sum_{j=1}^{n} \lambda_j y_j^t,\ L \le e\lambda \le U,\ \lambda \ge 0\right\},$$

where e is the row vector with all elements being equal to one and $\lambda \in R^n$ is the intensity vector, and L and U are the lower and upper bounds of the sum of the intensity respectively. $(L,U)=(0,\infty)$, $(1,1)$, $(1,\infty)$ and $(0,1)$ correspond to the CCR (Charnes-Cooper-Rhodes or Constant Returns to Scale = CRS), BCC (Banker-Charnes-Cooper or Variable Returns to Scale = VRS), IRS (Increasing Returns to Scale) and DRS (Decreasing Returns to Scale) models respectively. The production possibility set $(X,Y)^t$ is characterized by frontiers that are composed of $(x,y) \in (X,Y)^t$ such that it is not possible to improve any element of the input x or any element of the output y without worsening some other input or output. See Cooper et al. (1999) for further discussion on this and related subjects. We call this frontier set the *frontier technology* at the period t. In the Malmquist index analysis, the efficiencies of DMUs $(x_o, y_o)^1$ and $(x_o, y_o)^2$ are evaluated by the frontier technologies 1 and 2 in several ways.

2.1 Catch-up effect

The catch-up effect is measured by the following formula.

$$\text{Catch-up} = \frac{\text{Efficiency of } (x_o, y_o)^2 \text{ w.r.t. the period 2 frontier}}{\text{Efficiency of } (x_o, y_o)^1 \text{ w.r.t. the period 1 frontier}}. \qquad (8.1)$$

We evaluate catch-up of each DMU defined in the above formula by appropriate DEA models. A simple example in the case of a single input and output technology is illustrated in Figure 8-1.

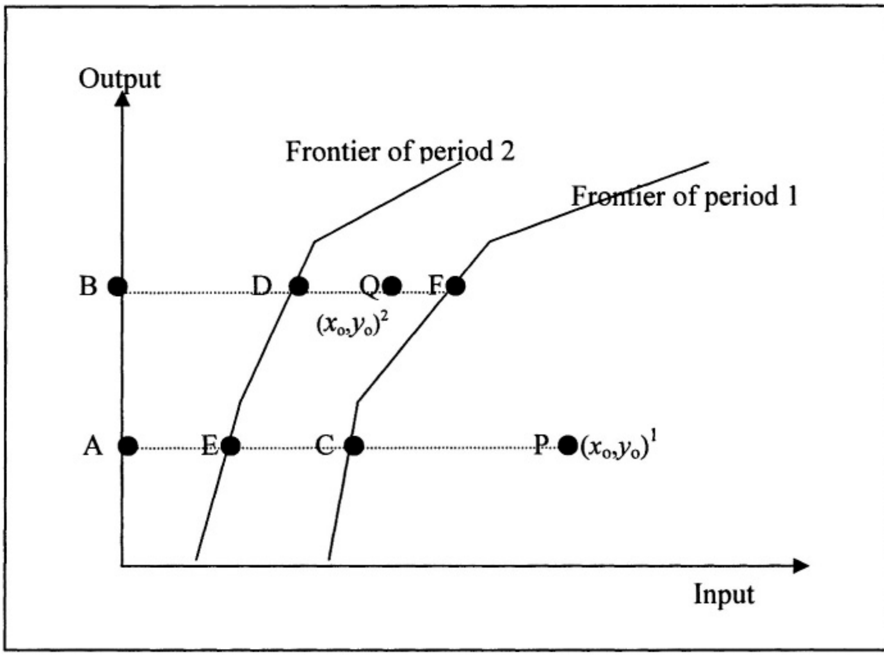

Figure8-1.: Catch-up

The catch-up effect (in input-orientation) can be computed as:

$$\text{Catch}-\text{up} = \frac{\text{BD}}{\text{BQ}} \Big/ \frac{\text{AC}}{\text{AP}}. \tag{8.2}$$

(Catch-up) > 1 indicates progress in the relative efficiency from period 1 to 2, while (Catch-up) = 1 and (Catch-up) < 1 indicate respectively no change and regress in efficiency.

2.2 Frontier-shift effect

In addition to the catch-up term, we must take account of the frontier-shift (innovation) effect in order to fully evaluate the productivity change since the catch-up effect is determined by the efficiencies being measured by the distances from the respective frontiers. In Figure 8-1 case, this can be explained as follows. The reference point C of $(\boldsymbol{x}_o, \boldsymbol{y}_o)^1$ moved to E on the frontier of period 2. Thus, the frontier-shift effect at $(\boldsymbol{x}_o, \boldsymbol{y}_o)^1$ is evaluated by

$$\varphi_1 = \frac{\text{AC}}{\text{AE}}. \tag{8.3}$$

This is equivalent to

$$\varphi_1 = \frac{\frac{AC}{AP}}{\frac{AE}{AP}} = \frac{\text{Efficiency of } (x_o, y_o)^1 \text{ w.r.t. the period 1 frontier}}{\text{Efficiency of } (x_o, y_o)^1 \text{ w.r.t. the period 2 frontier}}. \tag{8.4}$$

The numerator of the right hand side of (8.4) is already captured in (8.1). The denominator is measured as the efficiency score of $(x_o, y_o)^1$ relative to the period 2 frontier. Similarly, the frontier-shift effect at $(x_o, y_o)^2$ is expressed by

$$\varphi_2 = \frac{\frac{BF}{BQ}}{\frac{BD}{BQ}} = \frac{\text{Efficiency of } (x_o, y_o)^2 \text{ w.r.t. the period 1 frontier}}{\text{Efficiency of } (x_o, y_o)^2 \text{ w.r.t. the period 2 frontier}}. \tag{8.5}$$

Using φ_1 and φ_2, we define "Frontier-shift" effect by their geometric mean, i.e.,

$$\text{Frontier - shift} = \varphi = \sqrt{\varphi_1 \varphi_2}\,. \tag{8.6}$$

(Frontier-shift) > 1 indicates progress in the frontier technology around the DMU_o from period 1 to 2, while (Frontier-shift) = 1 and (Frontier-shift) < 1 indicate respectively the status quo and regress in the frontier technology.

2.3 Malmquist index

The Malmquist index (MI) is computed as the product of (Catch-up) and (Frontier-shift), i.e.,

$$\text{MI} = (\text{Catch - up}) \times (\text{Frontier - shift}). \tag{8.7}$$

We now employ the following notation for the efficiency score of DMU $(x_o, y_o)^{t_1}$ measured by the frontier technology t_2.

$$\delta^{t_2}((x_o, y_o)^{t_1}) \qquad (t_1 = 1, 2 \text{ and } t_2 = 1, 2). \tag{8.8}$$

Using this notation, the catch-up effect (C) in (8.1) can be expressed as

$$C = \frac{\delta^2((x_o, y_o)^2)}{\delta^1((x_o, y_o)^1)}. \tag{8.9}$$

The frontier-shift effect is described as

$$F = \left[\frac{\delta^1((x_o, y_o)^1)}{\delta^2((x_o, y_o)^1)} \times \frac{\delta^1((x_o, y_o)^2)}{\delta^2((x_o, y_o)^2)}\right]^{1/2}. \quad (8.10)$$

As the product of C and F, we obtain the following formula for the computation of MI, the Malmquist Index,

$$\mathrm{MI} = \left[\frac{\delta^1((x_o, y_o)^2)}{\delta^1((x_o, y_o)^1)} \times \frac{\delta^2((x_o, y_o)^2)}{\delta^2((x_o, y_o)^1)}\right]^{1/2}. \quad (8.11)$$

This last expression gives an another interpretation of MI, i.e., the geometric means of the two efficiency ratios: the one being the efficiency change measured by the period 1 technology and the other the efficiency change measured by the period 2 technology.

As can be seen from these formulas, the MI consists of four terms: $\delta^1((x_o, y_o)^1)$, $\delta^2((x_o, y_o)^2)$, $\delta^1((x_o, y_o)^2)$ and $\delta^2((x_o, y_o)^1)$. The first two are related to the measurements *within* the same time period, while the last two are for *intertemporal* comparison.

MI >1 indicates progress in the total factor productivity of the DMU_o from period 1 to 2, while MI = 1 and MI < 1 indicate respectively the status quo and decay in the total factor productivity.

3. MEASUREMENT OF MALMQUIST INDEX

In the non-parametric framework the Malmquist index (MI) is constructed by means of DEA technologies. There are a number of ways to compute MI. First, Färe, Grosskopf, Lindgren and Roos (1989, 1994) utilized the input and output oriented radial DEA model to compute MI. However, the radial model suffers from one shortcoming, i.e., neglect of slacks. To overcome this shortcoming, MI can be computed using the "slacks-based" non-radial and oriented DEA model. Alternatively MI can be computed from the non-radial and non-oriented DEA model.

3.1 The radial MI

The input-oriented radial MI measures the *within* and *intertemporal* scores by the linear programs given below.

[*Within* score in input-orientation]

$$\delta^s((x_o, y_o)^s) = \min_{\theta,\lambda} \theta \quad (8.12)$$
$$\text{subject to } \theta x_o^s \geq X^s \lambda$$
$$y_o^s \leq Y^s \lambda$$
$$L \leq e\lambda \leq U$$
$$\lambda \geq 0,$$

where $X^s = (x_1^s, \cdots, x_n^s)$ and $Y^s = (y_1^s, \cdots, y_n^s)$ are respectively input and output matrices (observed data) at the period *s*. We solve this program for *s* = 1 and 2. It holds that $\delta^s((x_o, y_o)^s) \leq 1$, and $\delta^s((x_o, y_o)^s) = 1$ indicates $(x_o, y_o)^s$ being on the technically efficient frontiers of $(X, Y)^s$.

[*Intertemporal* score in input-orientation]

$$\delta^s((x_o, y_o)^t) = \min_{\theta,\lambda} \theta \quad (8.13)$$
$$\text{subject to } \theta x_o^t \geq X^s \lambda$$
$$y_o^t \leq Y^s \lambda$$
$$L \leq e\lambda \leq U$$
$$\lambda \geq 0.$$

We solve this program for the pairs (*s*, *t*) = (1, 2) and (2, 1). If $(x_o, y_o)^t$ is not enveloped by the technology at the period *s*. the score $\delta^s((x_o, y_o)^t)$, if it exists, results in the value greater than 1. This corresponds to the concept of super-efficiency proposed by Andersen and Petersen (1993).

Although the above schemes are input-oriented, we can develop the output-oriented MI as well by means of the output-oriented radial DEA models. This is explained below.

[*Within* score in output-orientation]

$$\delta^s((x_o, y_o)^s) = \min_{\theta,\lambda} \theta \quad (8.14)$$
$$\text{subject to } x_o^s \geq X^s \lambda$$
$$(1/\theta)\, y_o^s \leq Y^s \lambda$$
$$L \leq e\lambda \leq U$$
$$\lambda \geq 0.$$

[*Intertemporal* score in output-orientation]

$$\delta^s((x_o, y_o)^t) = \min_{\theta,\lambda} \theta \quad (8.15)$$
$$\text{subject to } x_o^t \geq X^s \lambda$$
$$(1/\theta)\, y_o^t \leq Y^s \lambda$$
$$L \leq e\lambda \leq U$$
$$\lambda \geq 0.$$

[*Remark* 8.1: Inclusive or Exclusive scheme]

In evaluating the *within* score $\delta^s((x_o, y_o)^s)$, there are two schemes: one 'inclusive' and the other 'exclusive'. 'Inclusive' scheme means that, when we evaluate $(x_o, y_o)^s$ with respect to the technology $(X,Y)^s$, the DMU $(x_o, y_o)^s$ is always included in the evaluator $(X,Y)^s$, thus resulting in a score not greater than 1. 'Exclusive' scheme employs a method in which the DMU $(x_o, y_o)^s$ is removed from the evaluator group $(X,Y)^s$. This method of evaluation is equivalent to that of super-efficiency evaluation, and the score, if it exists, may be greater than 1. The *intertemporal* comparisons naturally utilize this 'exclusive' scheme. So adoption of this scheme even in the *within* evaluations is not unnatural and promotes the discrimination power. Refer to Section 5.2 for further discussions on this subject.

[*Remark* 8.2: Infeasible LP issues]

In the BCC (VRS) model [(*L*, *U*) = (1,1): variable returns to scale], it may occur that the intertemporal LP (8.13) has no solution in its input and output orientations. In the case of input-oriented model, (8.13) has no feasible solution if there exists i such that $y_{io}^t > \max_j \{y_{ij}^s\}$ whereas in the output-oriented case, (8.15) has no feasible solution if there exists i such that $x_{io}^t < \min_j \{x_{ij}^s\}$. In the IRS model $[(L, U) = (1, \infty)$: increasing returns to scale], it may occur that the output-oriented intertemporal LP has no solution, while the input-oriented case is always feasible. In case of DRS model [(*L*, *U*) = (0,1): decreasing returns to scale], it might be possible that the input-oriented (8.13) has no solution, while the output-oriented model is always feasible. However, the CRS (CCR) model does not suffer from such trouble in its intertemporal measurement. Refer to Seiford and Zhu (1999) for infeasibility issues of radial super-efficiency models. One solution for avoiding this difficulty is to assign 1 to the score, since we have no means to evaluate the DMU within the evaluator group.

3.2 The non-radial and slacks-based MI

As noted earlier, the radial approaches suffer from the neglect of slacks. In an effort to overcome this problem, several authors, e.g., Zhu (1996, 2001), Tone (2001, 2002) and Chen (2003) have developed non-radial measures of efficiency and super-efficiency. We develop here the non-radial and slacks-based MI.

First, we introduce the input-oriented SBM (slacks-based measure) and -Super-SBM (Tone (2001, 2002)). The SBM evaluates the efficiency of the examinee $(x_o, y_o)^s$ (s = 1,2) with respect to the evaluator set $(X,Y)^t$ (t = 1, 2) with help of the following LP:

[SBM-I]

$$\delta^t((x_o,y_o)^s)=\min_{\lambda,s^-}\ 1-\frac{1}{m}\sum_{i=1}^{m}s_i^-/x_{io}^s \tag{8.16}$$
$$\text{subject to } x_o^s=X^t\lambda+s^-$$
$$y_o^s\le Y^t\lambda$$
$$L\le e\lambda\le U$$
$$\lambda\ge 0,s^-\ge 0.$$

Or equivalently,
[SBM-I]

$$\delta^t((x_o,y_o)^s)=\min_{\theta,\lambda}\frac{1}{m}\sum_{i=1}^{m}\theta_i \tag{8.17}$$
$$\text{subject to } \theta_i x_{io}^s\ge\sum\lambda_j x_{ij}^t \quad (i=1,\cdots,m)$$
$$y_o^s\le Y^t\lambda$$
$$\theta_i\le 1\,(\forall i)$$
$$L\le e\lambda\le U$$
$$\lambda\ge 0,$$

where the vector $s^-\in R^m$ denotes input-slacks. The equivalence between (8.16) and (8.17) can be shown as follows: Let $\theta_i=(1-s_i^-/x_{io}^s)$. Then it holds that $\theta_i\le 1\,(\forall i)$ and the equivalence follows straightforward. This model takes input slacks (surpluses) into account but no output slacks (shortfalls). Notice that, under the 'inclusive' scheme (see *Remark* 8.2 above), [SBM-I] is always feasible for the case where $s=t$. However, under the 'exclusive' scheme, we remove $(x_o,y_o)^s$ from the evaluator group $(X,Y)^s$ and hence [SBM-I] may have no feasible solution even for the case $s=t$. In this case, we solve the [Super-SBM-I] below:

[Super-SBM-I]

$$\delta^t((x_o,y_o)^s)=\min_{\lambda,s^-}\ 1+\frac{1}{m}\sum_{i=1}^{m}s_i^-/x_{io}^s \tag{8.18}$$
$$\text{subject to } x_o^s\ge X^t\lambda-s^-$$
$$y_o^s\le Y^t\lambda$$
$$L\le e\lambda\le U$$
$$\lambda\ge 0,s^-\ge 0.$$

Or equivalently,
[Super-SBM-I]

$$\delta^t((x_o,y_o)^s)=\min_{\theta,\lambda}\frac{1}{m}\sum_{i=1}^{m}\theta_i \tag{8.19}$$
$$\text{subject to } \theta_i x_{io}^s\ge\sum\lambda_j x_{ij}^t \quad (i=1,\cdots,m)$$
$$y_o^s\le Y^t\lambda$$

$$\theta_i \geq 1 \ (\forall i)$$
$$L \leq e\lambda \leq U$$
$$\lambda \geq 0.$$

In this model, the score, if exists, satisfies $\delta^t((x_o, y_o)^s) \geq 1$.
In the output-oriented case, we solve the following LPs.

[SBM-O]

$$\delta^t((x_o, y_o)^s) = \min_{\lambda, s^+} 1 \Big/ \left(1 + \tfrac{1}{q} \sum_{i=1}^{q} s_i^+ / y_{io}^s\right) \tag{8.20}$$
$$\text{subject to } x_o^s \geq X^t \lambda$$
$$y_o^s = Y^t \lambda - s^+$$
$$L \leq e\lambda \leq U$$
$$\lambda \geq 0, s^+ \geq 0,$$

where the vector $s^+ \in R^q$ denotes the output-slacks.

Or equivalently,
[SBM-O]

$$\delta^t((x_o, y_o)^s) = \min_{\lambda, \phi} 1 \Big/ \left(\tfrac{1}{q} \sum_{i=1}^{q} \phi_i\right) \tag{8.21}$$
$$\text{subject to } x_o^s \geq X^t \lambda$$
$$\phi_i y_o^s \leq \sum_{j=1}^{n} y_{ij}^t \lambda_j \quad (i = 1, \ldots, q)$$
$$\phi_i \geq 1 \quad (\forall i)$$
$$L \leq e\lambda \leq U$$
$$\lambda \geq 0.$$

[Super-SBM-O]

$$\delta^t((x_o, y_o)^s) = \min_{\lambda, s^+} 1 \Big/ \left(1 - \tfrac{1}{q} \sum_{i=1}^{q} s_i^+ / y_{io}^s\right) \tag{8.22}$$
$$\text{subject to } x_o^s \geq X^t \lambda$$
$$y_o^s \leq Y^t \lambda + s^+$$
$$L \leq e\lambda \leq U$$
$$\lambda \geq 0, s^+ \geq 0.$$

Or equivalently,

[Super-SBM-O]

$$\delta^t((x_o, y_o)^s) = \min_{\lambda,\phi} 1 \Big/ \left(\tfrac{1}{q}\sum_{i=1}^{q} \phi_i\right) \tag{8.23}$$

$$\text{subject to } x_o^s \geq X^t\lambda$$

$$\phi_i y_o^s \leq \sum_{j=1}^{n} y_{ij}^t \lambda_j \quad (i = 1, \cdots, q)$$

$$0 \leq \phi_i \leq 1 \quad (\forall i)$$

$$L \leq e\lambda \leq U$$

$$\lambda \geq 0.$$

The output-oriented models take all output slacks (shortfalls) into account but no input slacks (surpluses).

The non-radial and slacks-based MI evaluates the four elements of MI: $\delta^1((x_o, y_o)^1)$, $\delta^2((x_o, y_o)^2)$, $\delta^1((x_o, y_o)^2)$ and $\delta^2((x_o, y_o)^1)$ by means of the LPs [SBM-I] and [Super-SBM-I]. This is now explained below.

3.2.1 The inclusive scheme case

1. The within scores $\delta^1((x_o, y_o)^1)$ and $\delta^2((x_o, y_o)^2)$ are measured by solving [SBM-I] or [SBM-O]. They are not greater than 1.
2. For the intertemporal scores $\delta^1((x_o, y_o)^2)$ and $\delta^2((x_o, y_o)^1)$, we solve the corresponding [SBM-I] or [SBM-O]. If they are feasible, the optimal value is the score that does not exceed 1. Otherwise, if they are infeasible, we apply [Super-SBM-I] or [Super-SBM-O] for measuring the score. In this case, the score, if exists, is always greater than 1.

3.2.2 The exclusive scheme case

1. To compute each component of MI, we apply [SBM-I] or [SBM-O] under the exclusive scheme. If they are feasible, the optimal value is the score that will never exceed 1.
2. Otherwise, if they are infeasible, we solve [Super-SBM-I] or [Super-SBM-O] to compute the score. In this case, the score, if exists, is always greater than 1.

[*Remark* 8.3: Another Non-radial model]

Chen (2003) proposed a non-radial Malmquist productivity index in which free slacks s^- in (8.14) are allowed. This model is also able to incorporate the decision maker's preference over performance improvement as introduced by Zhu (1996). Chen's model is described, in the non-preference case, as follows:

$$\delta^t((x_o, y_o)^s) = \min_{\theta,\lambda} \frac{1}{m}\sum_{i=1}^{m}\theta_i \tag{8.24}$$

$$\text{subject to } \theta_i x_{io}^s \geq \sum_{j=1}^{n}\lambda_j x_{ij}^t \qquad (i = 1, \cdots, m)$$

$$y_o^s \leq Y^t\lambda$$

$$\theta_i : \text{free } \forall i$$

$$\lambda \geq 0.$$

[SBM-I] in (8.17) and [Super-SBM-I] in (8.19) correspond to the cases with the additional conditions $\theta_i \leq 1 (\forall i)$ and $\theta_i \geq 1 (\forall i)$ to (8.24), respectively.

[*Remark* 8.4: Infeasible LP issues]
These models may suffer from the same infeasible troubles as the radial ones may encounter.

3.3 The non-radial and non-oriented MI

The models in this category deal with input and output slacks. The [SBM] and [Super-SBM] models used for computing $\delta^t((x_o, y_o)^s)$ are represented by the following fractional programs:

[SBM]

$$\delta^t((x_o, y_o)^s) = \min_{\lambda, s^-, s^+} \left(1 - \frac{1}{m}\sum_{i=1}^{m} s_i^- / x_{io}^s\right) \Big/ \left(1 + \frac{1}{q}\sum_{i=1}^{q} s_i^+ / y_{io}^s\right) \tag{8.25}$$

$$\text{subject to } x_o^s = X^t\lambda + s^-$$

$$y_o^s = Y^t\lambda - s^+$$

$$L \leq e\lambda \leq U$$

$$\lambda \geq 0, s^- \geq 0, s^+ \geq 0.$$

Or equivalently,
[SBM]

$$\delta^t((x_o, y_o)^s) = \min_{\theta,\phi,\lambda} \left(\frac{1}{m}\sum_{i=1}^{m}\theta_i\right) \Big/ \left(\frac{1}{q}\sum_{i=1}^{q}\phi_i\right) \tag{8.26}$$

$$\text{subject to } \theta_i x_{io}^s \geq \sum_{j=1}^{n} x_{ij}^t \lambda_j \quad (i = 1, \cdots, m)$$
$$\phi_i y_{io}^s \leq \sum_{j=1}^{n} y_{ij}^t \lambda_j \quad (i = 1, \cdots, q)$$
$$\theta_i \leq 1 \, (\forall i), \quad \phi_i \geq 1 \, (\forall i)$$
$$L \leq e\lambda \leq U$$
$$\lambda \geq 0.$$

[Super-SBM]

$$\delta^t((x_o, y_o)^s) = \min_{\lambda, s^-, s^+} \left(\frac{1}{m} \sum \bar{x}_i / x_{io}^s \right) \Big/ \left(\frac{1}{q} \sum_{i=1}^{q} \bar{y}_i / y_{io}^s \right) \qquad (8.27)$$

$$\text{subject to } \bar{x} \geq X^t \lambda$$
$$\bar{y} \leq Y^t \lambda$$
$$\bar{x} \geq x_o^s, \bar{y} \leq y_o^s$$
$$L \leq e\lambda \leq U$$
$$\bar{y} \geq 0, \lambda \geq 0.$$

Or equivalently,
[Super-SBM]

$$\delta^t((x_o, y_o)^s) = \min_{\theta, \phi, \lambda} \left(\frac{1}{m} \sum_{i=1}^{m} \theta_i \right) \Big/ \left(\frac{1}{q} \sum_{i=1}^{q} \phi_i \right) \qquad (8.28)$$
$$\text{subject to } \theta_i x_{io}^s \geq \sum_{j=1}^{n} x_{ij}^t \lambda_j \quad (i = 1, \cdots, m)$$
$$\phi_i y_{io}^s \leq \sum_{j=1}^{n} y_{ij}^t \lambda_j \quad (i = 1, \cdots, q)$$
$$\theta_i \geq 1 \, (\forall i), \quad 0 \leq \phi_i \leq 1 \, (\forall i)$$
$$L \leq e\lambda \leq U$$
$$\lambda \geq 0.$$

These fractional programs can be transformed into LPs. See Tone (2002) for detailed discussions. This model under the exclusive scheme (see *Remark* 8.1) evaluates the four components of MI: $\delta^1((x_0, y_0)^1)$, $\delta^2((x_o, y_o)^2)$, $\delta^1((x_o, y_o)^2)$ and $\delta^2((x_o, y_o)^1)$ using [SBM], and, if the corresponding LP is found infeasible, we then apply [Super-SBM].

[*Remark* 8.5: Infeasible LP issues]

For this non-oriented model, [Super-SBM] is always feasible and has a finite minimum in any RTS environment, but under some mild conditions, i.e., for each output i (= 1, ⋯, q) at least two DMUs have positive values. This can be seen from the constraints in (8.28). See Tone (2002) for the details.

4. SCALE EFFICIENCY CHANGE

Changes in efficiency may also be associated with returns to scale efficiencies. We therefore treat this topics as follows:

Let $\delta_C^{t_1}((x_o,y_o)^{t_2})$ and $\delta_V^{t_1}((x_o,y_o)^{t_2})$ denote the scores $\delta^{t_1}((x_o,y_o)^{t_2})$ (8.8) that are obtained under CRS and VRS environments respectively. It holds, for any combination of t_1 and t_2, that

$$\delta_C^{t_1}((x_o,y_o)^{t_2}) \le \delta_V^{t_1}((x_o,y_o)^{t_2}) \,. \tag{8.29}$$

We define the *scale efficiency* of $(x_o,y_o)^{t_2}$ relative to the technology $(X,Y)^{t_1}$ by

$$\sigma^{t_1}(x_o,y_o)^{t_2} = \delta_C^{t_1}((x_o,y_o)^{t_2})/\delta_V^{t_1}((x_o,y_o)^{t_2}). \tag{8.30}$$

From (8.29) it holds that $\sigma^{t_1}(x_o,y_o)^{t_2} \le 1$, and, if $\sigma^{t_1}(x_o,y_o)^{t_2} = 1$, it means that $(x_o,y_o)^{t_2}$ is positioned in the *most productive scale size* (MPSS) region of the technology $(X,Y)^{t_1}$, as named by Banker (1984) and Banker et al. (1984). Hence, $\sigma^{t_1}(x_o,y_o)^{t_2}$ is an efficiency indicator showing how far $(x_o,y_o)^{t_2}$ deviates from the point of technically optimal scale of operation. Several authors (Fare et al. (1994), Lovell and Grifell-Tatje (1994), Ray and Desli (1997), and Balk (2001)) have made attempts to enhance the Malmquist index by taking into account scale efficiency change effects. We introduce here two of them.

4.1 Proposal by Ray and Delsi (1997)

The Malmquist index in the CRS environment, MI_C, can be rewritten via the Malmquist index in VRS, MI_V, as follows:

$$MI_C = MI_V \times \left[\frac{\sigma^1(x_o,y_o)^2}{\sigma^1(x_o,y_o)^1} \times \frac{\sigma^2(x_o,y_o)^2}{\sigma^2(x_o,y_o)^1}\right]^{1/2} \tag{8.31}$$

Ray and Delsi interpret the term in the square parenthesis as the geometric mean of scale efficiency changes of (x_o, y_o) evaluated by the frotiers at the time periods 1 and 2. This can be illustrated in Figure 8-2 using the simple single-input and single-output case. The CRS frontier at the time period 1 is the dotted line (ray) through the origin as denoted by CRS^1, while the CRS frontier at 2 is the dotted line CRS^2. The VRS frontiers at 1(2) are the broken lines VRS^1 (VRS^2). The points $\text{P}(x_o, y_o)^1$ and $\text{Q}(x_o, y_o)^2$ designate the concerned DMU at 1 and 2 respectively. The numerator of the first fractional term in the parenthesis (8.31) above is (FI/FQ)/(FJ/FQ)=FI/FJ, while the denominator is (AE/AP)/(AC/AP)=AE/AC. Hence the first fractional term turns out to be (FI/FQ)/(AE/AC), i.e., the scale efficiency change from $\text{P}(x_o, y_o)^1$ to $\text{Q}(x_o, y_o)^2$ as evaluated by the $(X, Y)^1$ technology. Likewise, the second fractional term corresponds to the scale efficiency change evaluated by the $(X, Y)^2$ technology. Thus, their geometric mean indicates the average scale efficiency change of (x_o, y_o) from 1 to 2.

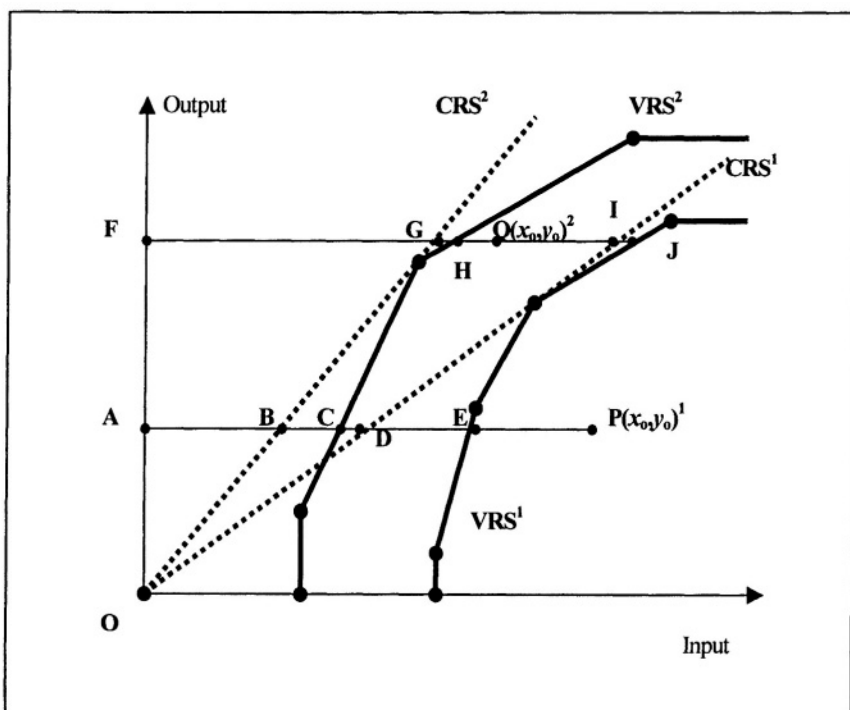

Figure 8-2. Scale efficiency change

Ray and Delsi decompose MI_C in the following manner.

$$\mathrm{MI}_C = \text{Catch - up}(V) \times \text{Frontier - shift}(V) \times \text{Scale efficiency change} \tag{8.32}$$

where

$$\text{Catch - up}(V) = \frac{\delta_V^2((x_o, y_o)^2)}{\delta_V^1((x_o, y_o)^1)} \tag{8.33}$$

$$\text{Frontier - shift}(V) = \left[\frac{\delta_V^1((x_o, y_o)^1)}{\delta_V^2((x_o, y_o)^1)} \times \frac{\delta_V^1((x_o, y_o)^2)}{\delta_V^2((x_o, y_o)^2)}\right]^{1/2} \tag{8.34}$$

and

$$\text{Scale - efficiency change} = \left[\frac{\sigma^1(x_o, y_o)^2}{\sigma^1(x_o, y_o)^1} \times \frac{\sigma^2(x_o, y_o)^2}{\sigma^2(x_o, y_o)^1}\right]^{1/2}. \tag{8.35}$$

4.2 Proposal by Balk (2001)

Balk (2001) introduced two fictitious DMUs $\mathbf{R} = (x_o^2, y_o^1)$ and $\mathbf{S} = (x_o^1, y_o^2)$ in addition to $\mathbf{P} = (x_o^1, y_o^1)$ and $\mathbf{Q} = (x_o^2, y_o^2)$. R (S) differs from P (Q) in the input-mix. The (input oriented) scale efficiency relative to the $(X,Y)^1$ technology is defined by

$$SEC^1(x_o^1, y_o^1, y_o^2) = \frac{\sigma^1(x_o^1, y_o^2)}{\sigma^1(x_o^1, y_o^1)}. \tag{8.36}$$

The numerator differs from Ray and Delsi's $\sigma^1(x_o^2, y_o^2)$. The rationale here is that, if this ratio is larger (smaller) than 1, the output combination y_o^2 lies closer to (farther away from) the point of technically optimal scale than y_o^1 did, distances being measured in the $x_o^1/\|x_o^1\|$-direction. Likewise, the (input-oriented) scale efficiency change relative to $(X,Y)^2$ technology is defined by

$$SEC^2(x_o^2, y_o^1, y_o^2) = \frac{\sigma^2(x_o^2, y_o^2)}{\sigma^2(x_o^2, y_o^1)}. \tag{8.37}$$

The scale efficiency change can be measured as the geometric mean of (8.36) and (8.37). Balk further introduced the *input-mix effect* by

$$ME^1(x_o^1, x_o^2, y_o^2) = \frac{\sigma^1(x_o^2, y_o^2)}{\sigma^1(x_o^1, y_o^2)}. \tag{8.38}$$

This measures how the relative distance of y_o^2 to the frontier of the CRS technology changes at period 1 when the input-mix changes. Similarly, the input-mix effect relative to the period 2 technology is measured by

$$ME^2(x_o^1, x_o^2, y_o^1) = \frac{\sigma^2(x_o^2, y_o^1)}{\sigma^2(x_o^1, y_o^1)}. \tag{8.39}$$

The product of these four factors, (8.36), (8.37), (8.38) and (8.39), results in the Ray and Delsi's scale-efficiency change (8.35). Hence, the latter is a combination of the scale efficiency change and the input-mix effect proposed by Balk. Eventually, Balk's decomposition turns out to be:

$$MI_C = MI_V \times \left[SEC^1(\cdot) \times SEC^2(\cdot)\right]^{1/2} \times \left[ME^1(\cdot) \times ME^2(\cdot)\right]^{1/2}. \tag{8.40}$$

5. ILLUSTRATIVE EXAMPLES FOR COMPARISONS OF MODELS

We here demonstrate and compare several models introduced in the preceding section using simple examples. The numerical results are obtained by using DEA-Solver Pro Version 4.0 (2003).

5.1 Radial vs. Non-radial

Table 8-1 exhibits a small example with two DMUs, A and B, each having two inputs x_1 and x_2, and a single output y, over two periods 1 and 2. At Period 1 DMU B has 1 unit of slacks in x_2 against A that is removed at Period 2.

For brevity, we use the notation $\delta^s(A^t)$ for designating the efficiency score of DMU A at Period t relative to the production frontier at Period s. The scores measured by the input-oriented radial CRS and the non-radial CRS models under the inclusive scheme are displayed in Table 8-2. The radial model records all scores as 1, whereas the non-radial (slacks-based) model returns $\delta^1(B^1)$ =0.75 and $\delta^2(B^1)$ =0.75 reflecting the existence of slacks in x_2 at Period 1.

Table 8-1. Example 1

	Period 1			Period 2		
DMU	x_1	x_2	y	x_1	x_2	y
A	1	1	1	1	1	1
B	1	2	1	1	1	1

Table 8-2. Input-oriented scores

	$\delta^1(A^1)$	$\delta^1(B^1)$	$\delta^1(A^2)$	$\delta^1(B^2)$	$\delta^2(A^1)$	$\delta^2(B^1)$	$\delta^2(A^2)$	$\delta^2(B^2)$
Radial	1	1	1	1	1	1	1	1
Non-radial	1	0.75	1	1	1	0.75	1	1

From Table 8-2, we obtained the catch-up (*C*), the frontier-shift (*F*) and the Malmquist index (MI) of each DMU from Period 1 to Period 2. These are recorded in Table 8-3.

Table 8-3. Catch-up, Frontier-shift and Malmquist index

	C(*A*)	*C*(*B*)	*F*(*A*)	*F*(*B*)	*MI*(*A*)	*MI*(*B*)
Radial	1	1	1	1	1	1
Non-radial	1	1.33	1	1	1	1.33

As can be seen from Table 8-1, DMU *B* has 1 unit of slacks in input x_2 against *A* at Period 1, while the slacks are removed at Period 2. This effort is well captured in the non-radial model (*C*(*B*)=1.33), but the radial model neglects this removal, as well as the existence, of the slacks. In consequence, DMU *B* is seen as improved in its productivity by 1.33 from Period 1 to Period 2 in the non-radial model, while the radial model reports no change in productivity, i.e., *MI*(*B*)=1.

5.2 Inclusive vs. Exclusive

In *Remark* 8.1, we noted that, 'inclusive' means that, when we evaluate $(x_o, y_o)^s$ with respect to the technology $(X,Y)^s$, the DMU $(x_o, y_o)^s$ is always included in the evaluator $(X,Y)^s$, thus resulting in the distance score being not greater than 1, while 'exclusive' employs the scheme that the DMU $(x_o, y_o)^s$ is removed from the evaluator group $(X,Y)^s$. Here we compare the two schemes using an example. Table 8-4 exhibits data set comprising three DMUs, *A*, *B* and *C*, with each one using a single input *x* to produce a single output *y*. Figure 8-3 shows the diagrammatical representation of this data set, where the legend A^1 indicates DMU *A* at the period 1.

Table 8-4. Example 2

	Period 1		Period 2	
	x	*y*	*x*	*y*
A	5	5	3	5
B	5	2	3	4
C	8	5	4	6

From the figure we can see that, under the CRS assumption, DMU *A* has advantage against *B* and *C* in both periods. However, the degree of superiority is less at the period 2 compared with that at period 1, implying that DMU *A* was *caught up* by its competitors.

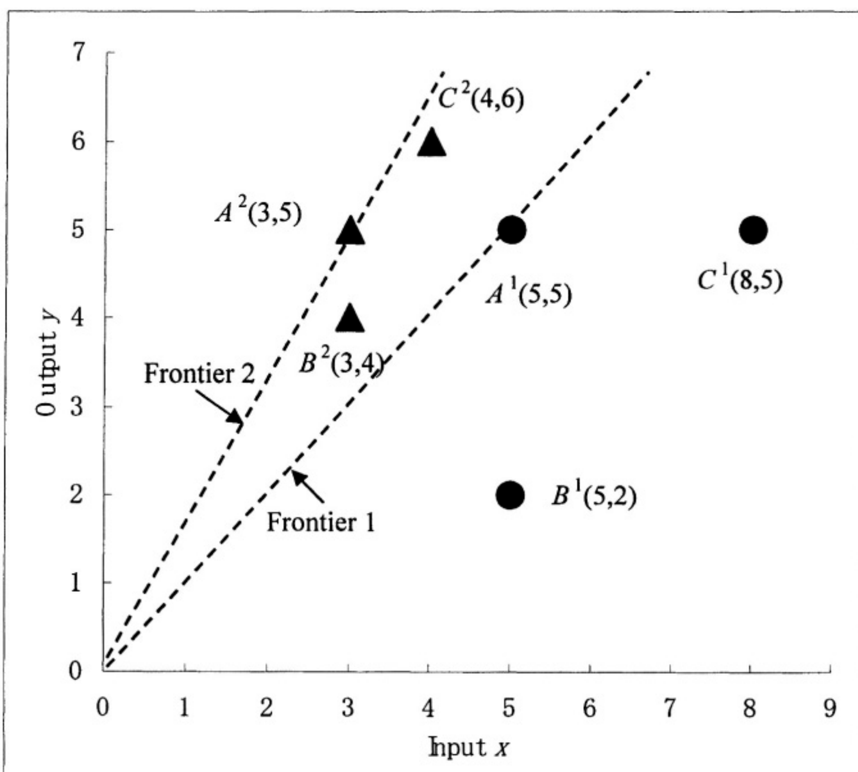

Figure 8-3. Three DMUs in two periods

First, we observe the catch-up effect of DMU *A*. The 'inclusive' CRS models assign 1 to both $\delta^1(A^1)$ and $\delta^2(A^2)$, since *A* is on the efficient frontier of the CRS models at both periods. Hence, there is no catch-up effect, i.e., $C(A)=1$. On the other hand, the input-oriented 'exclusive' non-radial CRS model evaluates A^1's superiority over B^1 and C^1 as $\delta^1(A^1)$ = 1+3/5 = 1.67 and A^2's superiority over B^2 and C^2 as $\delta^2(A^2)$= 1+0.33/3 = 1.11. Apparently, *A's* superiority is diminished in the period 2. The 'exclusive' model thus renders $C(A)$ = 1.11/1.67=0.69 which is less than 1. DMU *A* lost its superiority by about 40% at the period 2. Second, both the 'inclusive' and 'exclusive' CRS models yield the intertemporal distance scores: $\delta^1(A^2) = 1.67$ and $\delta^2(A^1) = 0.6$. Using the frontier-shift formula (8.10), we have $F(A)$ = 1.67 for the 'inclusive' model and $F(A)$ = 2 for the 'exclusive' model. The difference is due to those in $\delta^1(A^1)$ and $\delta^2(A^2)$ by the two schemes. So, we go further into the difference. We use the suffixes *incl* and *excl* for distinguishing the two schemes. Then, we have

$$\frac{F_{incl}(A)}{F_{excl}(A)} = \left[\left(\frac{\delta^1_{incl}(A^1)}{\delta^1_{excl}(A^1)}\right) \Big/ \left(\frac{\delta^2_{incl}(A^2)}{\delta^2_{excl}(A^2)}\right)\right]^{1/2} \tag{8.41}$$

Both the terms in the parenthesis on the right of (8.41) are not greater than 1, since it holds $\delta^s_{incl}(A^s) \le \delta^s_{excl}(A^s)$ for s = 1, 2. If the first term is greater (less) than the second, then we have $F_{incl}(A) \ge (\le) F_{excl}(A)$. In case of our example, $1.67 = F_{incl}(A) < F_{excl}(A) = 2$.

Apart from this example, let us observe each component on the right of (8.41). If A^1 is positioned in the relative interior of $(X,Y)^1$, then we have $\delta^1_{incl}(A^1) = \delta^1_{excl}(A^1) < 1$. Hence no difference arises between the two. If A^1 is positioned on the efficient frontier of $(X,Y)^1$, then we have $\delta^1_{incl}(A^1) = 1$ and $\delta^1_{excl}(A^1) \ge 1$. In this case, $\delta^1_{excl}(A^1)$ is measured as the distance from A^1 to the set $(X,Y)^1$ excluding A^1 that may be called the distance from A^1 to the *second* best frontier of $(X,Y)^1$ around A^1. Similarly, when A^2 is positioned on the efficient frontier of $(X,Y)^2$, then $\delta^2_{excl}(A^2)$ indicates the distance from A^2 to the second best frontier of $(X,Y)^2$ around A^2. Let us define $\alpha(\ge 1)$ and $\beta(\ge 1)$ by

$$\alpha = \frac{\delta^1_{excl}(A^1)}{\delta^1_{incl}(A^1)} \quad \text{and} \quad \beta = \frac{\delta^2_{excl}(A^2)}{\delta^2_{incl}(A^2)}$$

From (8.41), we have

$$F_{excl}(A) = F_{incl}(A) \times (\alpha / \beta)^{1/2} \tag{8.42}$$

If $\alpha / \beta > 1 (< 1)$, it means that *A*'s superiority is decreased (increased) against the second best frontier, and hence, the frontier-shift effect around *A* increased (decreased) from Period 1 to Period 2 compared to those under the inclusive scheme evaluation.

The Malmquist indexes under these two schemes are respectively obtained as

$$MI_{incl}(A) = C_{incl}(A) \times F_{incl}(A) = 1 \times 1.67 = 1.67 \text{ and}$$
$$MI_{excl}(A) = C_{excl}(A) \times F_{excl}(A) = 0.69 \times 2 = 1.38.$$

As can be seen, in the exclusive scheme, *A*'s superiority over *B* and *C* is lost while its frontier-shift effect is increased. So, the Malmquist index value in the exclusive scheme is lower for DMU *A* compared to that in inclusive scheme, indicating a good follow-up by competitors in Period 2.

5.3 Oriented vs. Non-oriented

Table 8-5 exhibits three DMUs, *A, B* and *C,* each having two inputs (x_1 and x_2) and two outputs (y_1 and y_2) for two time periods. As can be seen, DMU *A* outperforms *B* and *C* in Period 1 since DMU *B* has one unit of slacks in x_2 and DMU *C* one unit of slacks in y_1 against *A*. These slacks are removed in Period 2, thus enabling them to reach an even status.

Table 8-5. Example 3

	Period 1				Period 2			
	x_1	x_2	y_1	y_2	x_1	x_2	y_1	y_2
A	2	2	2	2	2	2	2	2
B	2	3	2	2	2	2	2	2
C	2	2	1	2	2	2	2	2

Using this data set, we compare four oriented models (input-oriented radial, input-oriented non-radial, output-oriented radial and output-oriented non-radial) and one non-oriented model (non-oriented non radial) under CRS environment. Table 8-6 displays the results. As expected, both radial models report no change as is seen from the second and fourth columns of this table, since all DMUs are on the efficient frontiers from the radial standpoint. The input-oriented non-radial model assigns 1.2 to *C*(*B*) reflecting the removal of input slacks of DMU *B* in x_2 at Period 2. However, this model gives a score of 1 (no change) to *C*(*C*), *F*(*C*) and *MI*(*C*) ignoring C's improvement in output y_1 at Period 2. The output-oriented non-radial model assigns a score of 1.5 to *C*(*C*), thus evaluating the removal of output shortfalls of *C* in y_1 at Period 2 but gives a score of 1 (no change) to *C*(*B*), *F*(*B*) and *MI*(*B*) neglecting *B*'*s* effort for removing input slacks in x_2. Finally, the non-oriented non-radial model takes account of all existing slacks, i.e., input surpluses and output shortfalls, and results in *MI*(*A*)=0.89, *MI*(*B*)=1.2 and *MI*(*C*)=1.5. As this example shows, the value of Malmquist index is conditional upon the model selection, i.e., radial or non-radial, or, oriented or non-oriented. So, due care is needed for deciding upon the selection of an appropriate model.

Table 8-6. Comparisons

	Input-oriented Radial	Input-oriented Non-radial	Output-oriented Radial	Output-oriented Non-radial	Non-oriented Non-radial
C(*A*)	1	0.8	1	0.75	0.8
C(*B*)	1	1.2	1	1	1.2
C(*C*)	1	1	1	1.5	1.5
F(*A*)	1	1.12	1	1.15	1.12
F(*B*)	1	1	1	1	1
F(*C*)	1	1	1	1	1
MI(*A*)	1	0.89	1	0.87	0.89
MI(*B*)	1	1.2	1	1	1.2
MI(*C*)	1	1	1	1.5	1.5

The legends *C*(*A*), *F*(*A*) and *MI*(*A*) denote the catch-up, frontier-shift and Malmquist index, respectively.

5.4 Infeasibility LP issues

As noted in *Remarks* 8.2 and 8.4, the oriented models in the VRS environment may suffer from the problem of infeasibility in measurement of the super-efficiency. However, the non-oriented VRS model is free from this trouble. We now demonstrate this fact with the help of a small data set reported in Table 8-7.

Table 8-7. Example 4

	Period 1		Period 2	
	x	*y*	*x*	*y*
A	3	1	1	3
B	4	2	2	4

In the oriented VRS model, we must solve the intertemporal scores $\delta^2(A^1)$, $\delta^2(B^1)$, $\delta^1(A^2)$ and $\delta^1(B^2)$ defined by (8.13), (8.15), (8.19) or (8.23). We encounter difficulties in solving LPs in this example. Specifically, in the input-oriented case, (8.13) and (8.19) have no feasible solution for $\delta^1(A^2)$ and $\delta^1(B^2)$, since $y_A^2 > \max\{y_A^1, y_B^1\}$ and $y_B^2 > \max\{y_A^1, y_B^1\}$. In the output-oriented case, (8.15) and (8.23) have no feasible solution for $\delta^1(A^2)$ and $\delta^1(B^2)$, since $x_A^2 < \min\{x_A^1, x_B^1\}$ and $x_B^2 < \min\{x_A^1, x_B^1\}$. Thus, in both cases, we cannot evaluate the frontier-shift effects and hence the Malmquist indexes are *not available* for all DMUs *A* and *B*. The non-oriented VRS model (8.28), however, can always measure these scores, as is seen from Table 8-8.

Table 8-8. Results by the non-oriented model

	$\delta^1(\cdot^1)$	$\delta^1(\cdot^2)$	$\delta^2(\cdot^1)$	$\delta^2(\cdot^2)$	Catch-up	Frontier	Malmquist
A	1.33	6	0.11	2	1.5	6	9
B	2	4	0.17	1.33	0.67	6	4

6. CONCLUDING REMARKS

In this chapter we have discussed Malmquist productivity index, and its measurements along with scale efficiency change and related subjects. All of the approaches we have covered in our opening paragraph utilize the ratio forms required in (8.11) with the exception of Thrall (2000) who provides an additive model. Thrall's formulation might best be regarded as an alternative to the MI and has therefore not been covered in this chapter. We conclude this chapter by pointing to several crucial issues involved in applying the Malmquist index model.

6.1 Selection of types of returns to scale

We need to exercise due care in selecting an appropriate RTS type. This identification problem is common to all DEA applications, since DEA is a data-oriented and non-parametric method. If the data set includes numeric values with a large difference in magnitude, e.g., comparing big companies with small ones, the VRS model may be a choice. However, if the data set consists of normalized numbers, e.g., per capita, acre and hour, the CRS model might be an appropriate candidate.

6.2 Radial or non-radial

The radial Malmquist index is based upon the radial DEA scores and hence remaining non-zero slacks are not counted in the scores. If slacks are not freely disposable, the radial Malmquist index cannot fully characterize the productivity change. In contrast, the input (output) oriented non-radial Malmquist index takes into account the input (output) slacks. This results in a smaller objective function value than in the radial model both in the *Within* and *Intertemporal* scores. Chen (2003) applied the non-radial Malmquist model to measure the productivity change of three major Chinese industries: Textiles, Chemicals and Metallurgicals. The result indicates that it is important to consider the possible non-zero slacks in measuring the productivity change. The radial model does not explain and hence cannot fully capture what the data say because of the large amount of slacks ignored.

6.3 Infeasibility and its solution

As pointed out in *Remarks* 8.2 and 8.4, the oriented VRS, IRS and DRS models suffer from the problem of infeasible solution in the *Intertemporal* score (super-efficiency) estimation. This is not a rare case, since the Malmquist model deals with data sets over time that display large variations in magnitude and hence the super-efficiency model suffers from many infeasible solutions. However, the non-radial and non-oriented model is free from such problems. In the radial framework, Bjurek (1996) has proposed the Malmquist total factor productivity index (MTFP) which computes an output-oriented index and an input-oriented index independently, and combines the two indexes into one. The MTFP is free from the infeasible problem in the VRS case.

6.4 Empirical studies

The Malmquist indexes have been used in a variety of studies. Färe et al. (1998a) surveyed empirical applications comprising various sectors such as agriculture, airlines, banking, electric utilities, insurance companies, and public sectors. However, most of these studies utilized the oriented-radial models. Thus, they suffer from the neglect of slacks and infeasibility. Empirical studies are at a nascent stage as of now.

REFERENCES

1. Andersen, P. and Petersen, N.C., 1993, A procedure for ranking efficient units in data envelopment analysis, *Management Science* 39, 1261-1264.
2. Balk, B.M., 2001, Scale efficiency and productivity change, *Journal of Productivity Analysis* 15, 159-183.
3. Banker, R.D., 1984, Estimating the most productive scale size using data envelopment analysis, *European Journal of Operational Research* 17, 35-44.

4. Banker, R.D., Charnes, A. and Cooper, W.W., 1984, Some models for estimating technical and scale inefficiencies in data envelopment analysis, *Management Science* 30, 1078-1092.
5. Bjurek, H., 1996, The Malmquist total factor productivity index, *Scandinavian Journal of Economics* 98, 303-313.
6. Caves, D.W., Christensen, L.R. and Diewert, W.E., 1982, The economic theory of index numbers and the measurement of input, output and productivity, *Econometrica* 50, 1393-1414,
7. Chen, Y., 2003, Non-radial Malmquist productivity index with an illustrative application to Chinese major industries, *International Journal of Production Economics* 83, No. 1, 27-35.
8. Cooper, W.W., Seiford, M.L. and Tone, K., 1999, *Data Envelopment Analysis: A Comprehensive Text with Models, Applications, References and DEA-Solver Software,* Boston, Kluwer Academic Publishers.
9. DEA-Solver Pro Version 4.0, 2003, Saitech-Inc, www.saitech-inc.com.
10. Färe, R. and Grosskopf, S., 1992, Malmquist indexes and Fisher ideal indexes, *The Economic Journal* 102, 158-160.
11. Färe, R., Grosskopf, S, Lindgren, B. and Roos, P., 1989, 1994, Productivity change in Swedish hospitals: a Malmquist output index approach, in Charnes, A., Cooper, W.W., Lewin, A.Y. and Seiford, M.L. (eds.) *Data Envelopment Analysis: Theory, Methodology and Applications,* Boston, Kluwer Academic Publishers.
12. Färe, R., Grosskopf, S, Norris, M. and Zhang, Z., 1994, Productivity growth, technical progress, and efficiency change in industrialized countries, *The American Economic Review* 84, 66-83.
13. Färe, R., Grosskopf, S. and Roos, P., 1998a, Malmquist productivity indexes: a survey of theory and practice, in *Index Numbers: Essays in Honour of Sten Malmquist*, Kluwer Academic Publishers, 127-190.
14. Färe, R., Grosskopf, S. and Russell, R., 1998b, *Index Numbers: Essays in Honour of Sten Malmquist,* Kluwer Academic Publishers.
15. Lovell, C.A.K. and Grifell-Tatje, E., 1994, A generalized Malmquist productivity index, Paper presented at the Georgia Productivity Workshop at Athens, GA, October 1994.
16. Malmquist, S., 1953, Index numbers and indifference surfaces, *Trabajos de Estadistica* 4, 209-242.
17. Ray, S.C. and Delsi, E., 1997, Productivity growth, technical progress, and efficiency change in industrialized countries: comment, *The American Economic Review* 87, 1033-1039.
18. Seiford, M.L. and Zhu, J., 1999, Infeasibility of super-efficiency data envelopment analysis, *INFOR* 37, 174-187.
19. Thrall, R.M., 2000, Measures in DEA with an application to the Malmquist index, *Journal of Productivity Analysis* 13, 125-137.

20. Tone, K., 2001, A slacks-based measure of efficiency in data envelopment analysis, *European Journal of Operational Research* 130, 498-509.
21. Tone, K., 2002, A slacks-based measure of super-efficiency in data envelopment analysis, *European Journal of Operational Research* 143, 32-41.
22. Zhu, J., 1996, Data envelopment analysis with preference structure, *Journal of the Operational Research Society* 47, 136-150.

Acknowledgements

The author would like to extend his sincere gratitude to the Editors and Biresh K Sahoo for their careful reading of the manuscript.